# Project Design Phase

## Solution Architecture

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|---|---|
| **Date** | 13 February 2026 |
| **Project id** | LTVIP2026TMIDS36768 |
| **Project name** | Plugging into the future :An exploration of data analyticals with tableau |
| **Minimum marks** | **4 marks** |

**Solution Architecture:**

**<u>1.Data Sources Layer:</u>**

Collect electricity consumption data from:

- Smart meters (household usage)
- Industrial meters
- Weather data APIs
- Historical electricity records
- Grid load data

Data Types:

- Timestamp
- Units consumed (kWh)
- Location
- Consumer type
- Temperature
- Peak/Off-peak indicators

**<u>2.Data Ingestion Layer</u>**

This layer collects and transfers data into the system.

Tools Used:

- API integration
- IoT gateways
- Batch uploads (CSV/Excel)
- Streaming ingestion tools

Real-time ingestion tools:

- Apache Kafka
- Apache NiFi

Purpose:

- Ensure continuous data flow
- Handle real-time smart meter data

**3.Data Storage Layer**

Stores both raw and processed data.

Storage Options:

- Cloud Data Lake
- Data Warehouse
- Example platforms:
- Amazon S3
- Google BigQuery
- Microsoft Azure SQL Database

Data Storage Types:

- Raw Data (Unprocessed)
- Cleaned Data
- Aggregated Data

**4.Data Processing & Analytics Layer**

This is the core intelligence layer.

Data Cleaning:

- Remove missing values
- Handle anomalies
- Normalize data

Data Transformation:

- Daily/Weekly/Monthly aggregation
- Peak hour detection
- Consumer segmentation

Advanced Analytics:

- Trend Analysis
- Load Forecasting
- Seasonal Pattern Detection
- Demand Prediction using ML models

Tools:

- Apache Spark
- Python
- TensorFlow

**5. Visualization & Reporting Layer**

Transforms insights into interactive dashboards.

Tools:

- Power BI
- Tableau

Dashboards show:

- Daily usage trends
- Monthly growth patterns
- Peak load analysis
- Regional consumption comparison
- Future demand prediction

**example:**

Vijayawada Smart City Electricity Monitoring

- 10,000 homes have smart meters
- Data collected every 15 minutes

System finds:

- High AC usage in summer
- Peak load at night

Insight:

Government introduces:

- Time-based electricity pricing
- Energy-saving awareness programs

Result:

- 15% reduction in peak load
- Better power grid stability

## Example:Solution Architecture

# Plugging into the Future:

## An Exploration of Electricity Consumption Patterns